MILITARY SPECIFICATION

WARHEAD, GUIDED MISSILE, HE, M207E1
LINER FOR

This Amendment forms a part of Military Specification MIL-W-63389 (AR), dated 7 January 1980, and is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1 DRAWINGS:

Delete "(SEE 6.4)"
Change title of drawing 9332524 from "Liner" to "Liner, Improved TOW" (Make this change in title wherever reference to drawing 9332524 appears)

* Add 2.2 as follows:

2.2 Other publications The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM B 248 General Requirements For Wrought Copper and Copper - Alloy Plate, Sheet, Strip, and Rolled Bar

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19102)

PAGE 2

Add 3.5 as follows:

"3.5 Data Requirements: The contractor shall generate data in accordance with requirements of data item description cited in 6.4."

FSC: 1336

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* Add 3.6, 3.6.1 and 3.6.2 as follows:

3.6 Process Control

3.6.1 Process Documentation. Concurrent with submission of the First Article Sample (see paragraph 4.3), the contractor shall submit a written description of the heat treatment process used to produce the samples together with the means by which the process can be verified in production. The method of verification shall be reviewed by and must be acceptable to the government. See Data Item Description DI-P-1604 (Tailored). Proposed changes to the process which produced the accepted First Article Sample shall be submitted to the government prior to actual implementation of the change. A new First Article Sample will be required for each process change.

3.6.2 Liner Traceability. The contractor shall also submit a method to establish liner traceability with specific annealed batches. This information shall be furnished as a part of DI-P-1604 (Tailored) above and shall be reviewed by and must be acceptable to the government.

PAGE 3

* 4.3.1 Under "Liner" add the following:

Sample material of the starting stock shall be selected prior to fabrication and used for hardness, chemical composition and grain size tests. In addition, ten finish machined liners from a single annealed batch shall be used for testing for grain size and hardness.

PAGE 4

* Delete Page 4 and substitute Page 4 attached.

PAGES 6 & 6A

* Delete Pages 6 and 6A and substitute Pages 6, 6A and 6B attached.

PAGE 7

* Add 4.4.3.1 & 4.4.3.1.1 as follows:

4.4.3.1 Heat Treatment. Tests shall be performed in accordance with 4.5.2.4 and 4.5.2.5.

4.4.3.1.1 Sample testing shall be in accordance with the following:

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a. Randomly select two liners per annealed batch and perform the tests in accordance with 4.5.2.4 and 4.5.2.5. An annealed batch shall consist of that quantity of liners that has been annealed in the same heat treat equipment at the same time by the same method and temperature in one unchanged process.

b. If five consecutive annealed batches meet the heat treatment requirements, reduce the sampling to one liner per month's production of annealed batches. Test results must be submitted to and be accepted by the government.

c. If an unacceptable liner is found, start again at step a. In addition, sample one liner from each annealed batch processed after the last acceptable quantity. Batches represented by unacceptable liners shall be rejected.

d. In addition to the sample test results, evidence must be submitted to establish that the heat treatment process used for each month's production is unchanged from the approved process (See para. 3.6.1). This evidence shall be in the form of permanent hard records, such as temperature-time tracings.

* Add 4.5.2, 4.5.2.1, 4.5.2.2, 4.5.2.3, 4.5.2.4 and 4.5.2.5 as follows:

4.5.2 Metallurgical & Chemical Tests The following tests shall be performed using equipment and procedures prescribed by ASTM B 248.

4.5.2.1 Hardness of starting stock. The Rockwell hardness of a sample of starting stock shall be evaluated. These test data shall be independent from the supplier data. Failure of the sample to comply with the requirements of ASTM B 152 shall cause rejection of that quantity of starting stock represented by the sample. The hardness test data shall be submitted in accordance with DI-T-5247 (Tailored) (See 6.4)

4.5.2.2 Chemical composition of starting stock. Sample of starting stock shall be analyzed for chemical requirements. These test data shall be independent from supplier data. Failure of the sample to comply with the requirements of ASTM B 152 shall cause rejection of that quantity of starting stock represented by the sample. Results of these tests shall be submitted in accordance with DI-T-5247 (Tailored) (See 6.4)

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4.5.2.3 Grain size of starting stock. Sample of starting stock shall be tested for grain size. This test shall be independent from supplier data. Failure of the sample to comply with the requirements of ASTM B 152 shall cause rejection of that quantity of starting stock represented by the sample. The grain size test data shall be submitted in accordance with DI-T-5247 (Tailored) (See 6.4)

4.5.2.4 Hardness of finished liner. The sample liners shall be sectioned longitudinally into four equal parts and marked for exact location relative to the finished liner. The Rockwell F hardness readings shall be taken one-inch from the base and every one inch thereafter in a longitudinal direction, from base to apex, for six readings on each section. (See Fig. 1). A total of twenty-four readings shall be taken for the finished liners. The hardness readings shall be identified as to their location on the liner. The recorded readings shall be analyzed for variations in hardness in both the longitudinal and circumferential directions. If any of the sample liners do not meet the hardness requirement specified on the drawing, the liner heat treat batch represented by the samples shall be rejected. The hardness test data shall be submitted in accordance with DI-T-5247 (Tailored) (See 6.4). This is a destructive test.

4.5.2.5 Grain size of finished liner. A test section from 4.5.2.4 shall be utilized for taking grain size. Grain size shall be taken on one of the quartered sections in a longitudinal direction 1 1/2-inch from the apex and 1 1/2-inch apart from each other for a total of four readings per sample liner. (See Fig. 1). If any of the sample liners do not meet the grain size requirements as specified on the drawing, the liner heat treat batch represented by the sample shall be rejected. The grain size test data shall be submitted in accordance with DI-T-5247 (Tailored) (See 6.4).

* Delete 6.4 and 6.5 in their entirety and substitute the following:

6.4: Data requirements - Deliverable data required by the specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.5	Ammunition Data Card	DI-L-1410
3.6, 3.6.1, & 3.6.2	Manufacturing Methods Report	DI-P-1604 (Tailored)
4.5.2, 4.5.3, 4.5.4, 4.5.5, & 4.5.6	Test Report	DI-T-5247 (Tailored)

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The above data items shall be delivered to the following:

Commander
U. S. Army Research and Development Command
ATTN: DRDAR-QAT-B
Dover, N. J., 07801

NOTE: The margins of this amendment are marked with an asterisk or vertical line to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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(Project 1336-A 468)

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Liner, Improved TOW		SHEET 1 OF 1		DRAWING NUMBER: 9332524
					NEXT HIGHER ASSEMBLY: 9332555
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Liner, Improved TOW (Dwg. 9332524)				
	Examination for Defects	20		3.2	4.4.2.1
	Hardness of Starting Stock	Sample Material		3.2	4.5.2.1
	Chemical Composition of Starting Stock	Sample Material		3.2	4.5.2.2
	Grain Size of Starting Stock	Sample Material		3.2	4.5.2.3
	Hardness of Finished Liner	10 (See Note 1)		3.2	4.5.2.4
	Grain Size of Finished Liner	10 (See Note 1)		3.2	4.5.2..5

NOTES:

1. The same 10 liners are used for both tests.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Liner, Improved TOW	1 OF 3	9332524
			NEXT HIGHER ASSEMBLY 9332555

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Runout of base with datums A & B		0.25%	3.2	Gage
102	Length from base to datum B		0.40%	3.2	Gage
103	Length from datum A to datum B		0.40%	3.2	Gage
104	Straightness on exterior surface		0.25%	3.2	Gage
105	Wall thickness variation in any axial plane		0.25%	3.2	Gage
106	Wall thickness variation in any transverse plane (See Note 2)		100%	3.2	Gage
107	Length from datum B to inside spherical radius		0.40%	3.2	Gage
108	Length from datum B to basic dimension at apex end		0.40%	3.2	Gage
109	Length from large to small basic dimensions at apex end		0.40%	3.2	Gage
110	Wall thickness at large basic dimension at apex end		0.25%	3.2	Gage
111	Wall thickness at small basic dimension at apex end		0.25%	3.2	Gage
112	Outside diameter		0.25%	3.2	Gage
113	Runout of outside diameter with datums A & B		0.40%	3.2	Gage

NOTES:

2. Measurements shall be taken on both 42° and 28° angle surfaces.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.1	Liner, Improved TOW		9332524
			NEXT HIGHER ASSEMBLY
			9332555

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined (Cont'd)				
114	Runout between two basic dimensions on outside surface at apex end (See Note 8 on Dwg.)		0.25%	3.2	Gage
115	Wall thickness at apex end		0.40%	3.2	Gage
116	Inside diameter		0.40%	3.2	Gage
117	Length of liner step		0.40%	3.2	Gage
118	Wall thickness in area at 3.95 dim.		0.40%	3.2	Gage
119	Evidence of discontinuity of machined surface		100%	3.2	Visual
120	Hardness of starting stock (See Note 3)	Mat'l Sample		3.2	4.5.2.1
121	Chemical Composition of Starting Stock (See Note 3)	Mat'l Sample		3.2	4.5.2.2
122	Grain Size of Starting Stock (See Note 3)	Mat'l Sample		3.2	4.5.2.3
123	Hardness of finished liner (See Note 3)	4.4.3.1& 4.4.3.1.1		3.2	4.5.2.4
124	Grain size of finished liner (See Note 3)	4.4.3.1& 4.4.3.1		3.2	4.5.2.5

NOTES:

3. Data Item Description DI-T-5247 (Tailored) Applies for these tests.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.1	Liner, Improved TOW				9332524
					NEXT HIGHER ASSEMBLY 9332555
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor	(Cont'd)				
201	Surface finish improper		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.4	Visual
203	Presence of four slots on O.D.		0.65%	3.2	Visual
204	Inside spherical radius, apex (See Note 1)		0.65%	3.2	Gage
205	Outside spherical radius, apex		0.65%	3.2	Gage

NOTE: 1. All new and resharpened tooling shall be certified prior to use. A cast of the inside spherical radius of the liner shall be made from "dental plaster" or equivalent, which shall be inspected by an optical comparator of 10X magnification or greater. This inspection shall be performed at the beginning of the shift, halfway through the shift and whenever tooling is changed.

NOTES:

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SCHEMATIC OF SECTIONED LINER

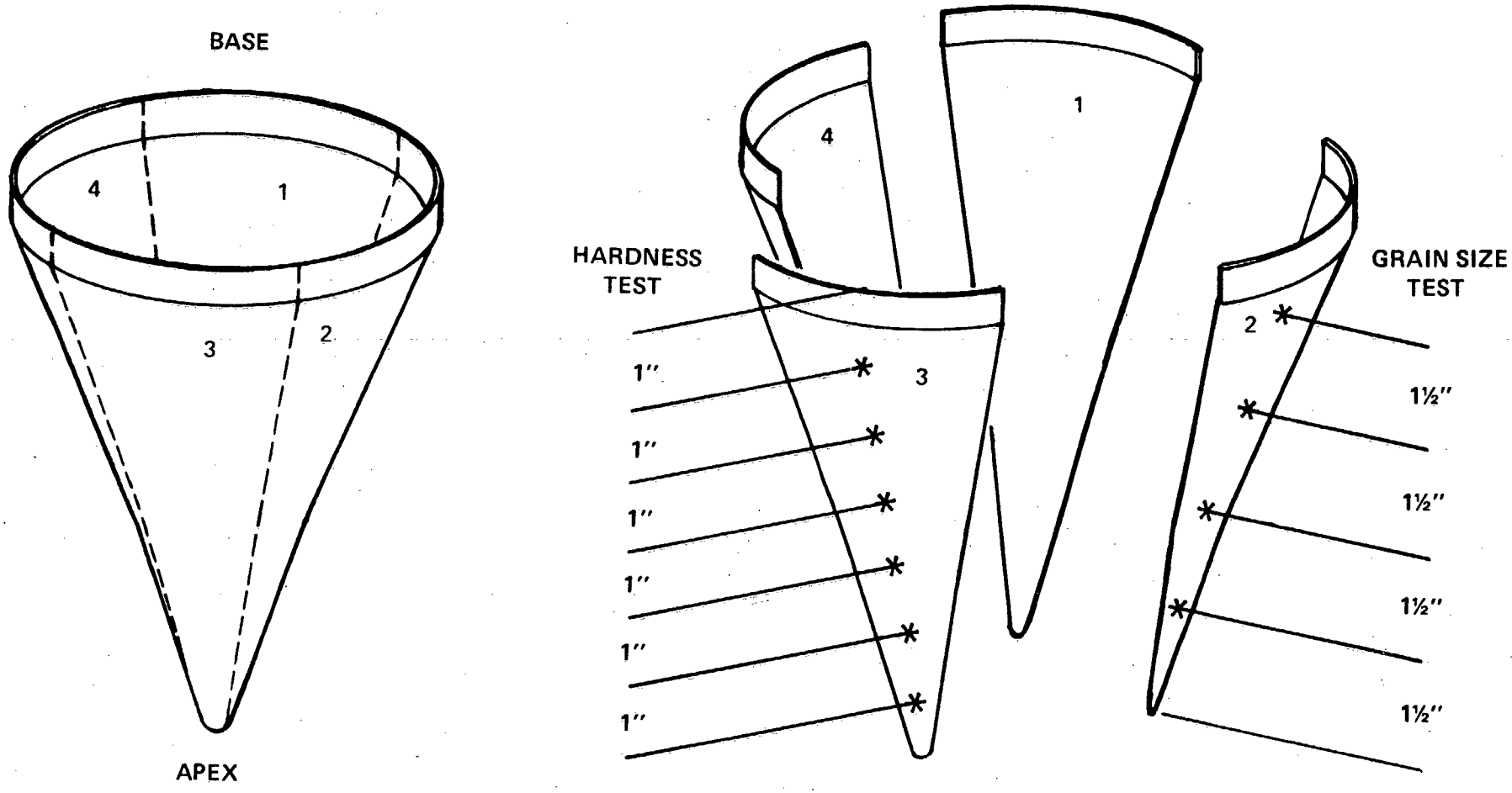


FIG. 1

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